# Configuring The User: the case of usability trials

Steve Woolgar
Centre for Research in Innovation Culture and Technology (CRICT)
Brunel University
Uxbridge
Middlesex UB8 3PH

Paper for discussion at Discourse Analysis Workshop, University of Lancaster, 25–26 September 1990.

Draft 1.1 Configur. doc
Please do not quote this preliminary draft without permission (1)

# Configuring the user: the case of usability trials

*Steve Woolgar*

How are you getting on with your paper for the Lancaster meeting?[2]

It's a mess. The usual thing – not enough time to really sort out the issues before having to send it off. I keep changing my mind about what are the key things to put in it, about what the key points of the argument are. So there are still lots of bugs in it, the analysis isn't finished . . .

I know. You'd like to make it better, but you've got to send it off. So I suppose you've put the standard warning on the front cover?

Right! 'Please do not quote this preliminary draft without permission'.[3]

Interesting how people do that. It's as if you're trying to delimit the extent of access to your text. On the one hand, writers use various devices to *attract* readers into the text. For example, The Abstract is a standard synechdotal device for suggesting the reader might profit from a closer relationship with the text. Where is The Abstract, by the way?

It's just there on the next page.

Oh, right. And on the other hand, writers often try to *control* the relationship between reader and text. So you encourage readers to relate to the text, but specify constraints on how they can use it. They can look, but not touch, so to speak.

Yes. It's a bit like that American advert for tamper-proof medicines which are sealed in cellophane: 'Shrink-wrapped for

your protection'. In that case, access to the product is specified by way of a guarantee that nobody else has been there before you. Or like those warnings on the disks when you buy a software package: 'By opening this sealed disk package, you are agreeing to become bound by the terms of the Microsoft License Agreement'. The manufacturers are saying that you should only access their text if you agree to use it in certain prescribed ways.[4]

So, do you think your paper is going to work?

I'm not sure. It depends on the audience. I don't really have a good sense of what reaction to expect. That's what so helpful about these Discourse Analysis Workshops. You get a chance to try out the paper before finally going public. It gives you the opportunity to figure out who the readers will be.

**Abstract**

The paper explores the metaphor of machine as text, set within the context of longstanding problems in social theory about agency and object. These problems concern both the conventional basis of attribution of intentionality and presumptions about the boundedness of entities. In particular, our preconceptions about the nature and capacity of different entities shape what counts as legitimate accounts of action and behaviour. Materials from an ethnography of computers are used to show how the design and production of a new entity (a new range of microcomputers) amounts to a process of configuring its user, where 'configuring' includes defining the identity of putative users, and setting constraints upon their likely future actions. Configuring occurs in a context where knowledge and expertise about users is socially distributed. As a result of this process, the new machine becomes its relationship with its configured users. In participants' determinations of the character of users, the new machine's case provides a powerful symbol of the boundary between insiders and outsiders to the company. An analysis of audio and video records of usability trials suggests the importance of boundary work in deciding the adequacy of the relationship between machine and user.

## I Introduction

It should be clear by now that technology, and information technology (IT) in particular, is just the latest excuse for doing social science. This is obvious in the prosaic sense that increased funds are available for social scientists (and others) who can construe their work as an attempt to situate technology within its social and cultural contexts. However, this is more than just a cynical comment about funding opportunism. The fact that IT has become a legitimate focus for research council funding reflects its emergence as a significant social phenomenon.[5] This, in turn, reflects a widespread perception of the potential effects of this new technology. The current mood displays concern about the extent of these effects and much effort is expended in determining the nature and capacity of IT.

This paper is part of a more general project which attempts to capitalise on current concerns about the nature of IT in order to address some longstanding questions in social theory and in the philosophy of social science, notably questions about agency and object. In essence, this project explores these issues by addressing questions about interpretation of technology. For example, it looks at how technology is understood vis-à-vis the actions and capability of humans and other animate entities; it attempts to understand how our ideas about the differences between animate and inanimate entities are modified with the introduction of new technology. Above all, it investigates the structure of a discourse which sustains and displays basic assumptions about the differential capacities of the entities with which we populate our world.

My strategy for attempting to realise these grandiose ambitions is the exploration of a metaphor: the machine as text.[6] The idea is to begin with the supposition that the nature and capacity of the machine is, at least in principle, interpretively flexible. This then sets the frame for an examination of the processes of construction (writing) and use (reading) of the machine; the relation between readers and writers is understood as mediated by the machine and by interpretations of what the machine is, what it's for, what it can do. To suggest that machines are texts is, of course, to deconstruct definitive versions of what machines can do. There is thus a sense in which the exploration of this metaphor challenges intuitive beliefs about technology; the 'actual' effects of technology are usually plain to see, and often brutishly incontrovertible. At the

same time, then, the exploration of the machine text metaphor deals with a particularly hard case in interpretation. Precisely because it is counter-intuitive to think of a machine as a text, this case might provide insights into more general questions about textuality.

It is worth stressing that the idea is to *explore* the metaphor, rather than merely to apply it. I have no wish to insist that machines *actually are* texts. Rather the point is to *play against* this metaphor, to see how far we can go with it.[7] What happens to the structure of our discourse when we introduce the notion of machine as text? What, if anything, is special about machines by comparison with other texts? What are the limits of talking in this bizarre way?

This paper is a preliminary attempt to play against one specific aspect of the machine text metaphor: the notion of the reader as user. As writers like Friedman (1989) have pointed out, the 1980s have seen considerable attention devoted to 'the problem of the user' amongst the designers and builders of computer systems. This paper takes the line that the emergence of a new range of microcomputers crucially entails the definition, delineation and emergence of The User. We could say that this process amounts to the (social) construction of the user. However, it is not just the identity of the user which is constructed. For along with negotiations over who the user might be, comes a set of design (and other) activities which attempt to define and delimit the user's possible actions. Consequently, it is better to say that by setting parameters for the user's actions, the evolving machine effectively attempts to *configure* the user.

The paper begins by outlining some problems of agency and attribution. A framework for addressing these problems is set out by way of a brief description of an eighteen month participant observation study carried out in a company which manufactures microcomputers. For reasons elaborated below, it is useful to construe this empirical study as an Ethnography of Computers. Particular attention is then given to a study of the 'usability trials' carried out by the User Products section of the company towards the latter stages of the project.

## II Agency and attribution

### *II.1 The conventional basis of attribution*

Some longstanding questions about human nature, action and agency, boil down to what we presume different entities (actors, agents) to be capable of. Our answers to this question, in turn, have consequences for what we take to be legitimate accounts of action and behaviour. For example, it is in virtue of our willingness to ascribe intentionality to certain (classes of) animate agent that we ask whether or not the armed robber *meant* to fire the gun. By contrast, our unwillingness to ascribe intentionality to certain inanimate entities makes other queries – did the gun want to fire? did it agree to being fired? – appear nonsensical or, at best, 'merely metaphorical', a 'literary device', the use of 'poetic licence'.

Of course, my use of 'our willingness' does not mean that ascription of intentionality is merely a matter of whim. If one day we felt like granting intentionality to an inanimate object – OK, let's say the gun wanted to fire! – this would not change much. Rather, the point is that the distribution of attributes is institutionalised in conventional practice. In other words, the distribution of attributes is sanctioned in virtue of conventions for correct/normal usage. Our 'unwillingness' to talk about guns wanting something reflects what we experience as awkwardnesses which arise when we try to describe things that way. A similar awkwardness occurs when we try to apply pronomial and behavioural characteristics to entities which are not normally credited with such. For example, 'the machine *who* thinks' (rather than the machine *that* thinks) invites comment about what novel claim we are making for a (particular) machine or suggests that some (merely) metaphorical usage is being employed.

Of course, things are much more complicated than this simple picture suggests. For a start, intentionality is not a quality ascribed uniformly to all types of animate agent. Amongst humans, for example, we are sometimes less likely to grant this quality to the very young, or to the old or to the mentally ill. There also appears to be a hierarchy of animate non-humans who (which?) more or less deserve attributions of cognitive states. (Dogs can be said to be looking happy, but gerbils?) More important, perhaps, is the point that there is no straightforward division of appropriate

mental predicates between animate and inanimate entities. We cannot therefore unproblematically map sets of predicates on to types of entity, precisely because predicates can be used to 'bring out' or 'suggest' or 'speak metaphorically about' the animate qualities of inanimate entities, and vice versa. For example, record players, cars and computers can be said to be temperamental. Space shuttles, at least until Challenger, can be said to be 'behaving perfectly', and so on. On the other side of the coin, animate (human) miscreants are sometimes said to have just behaved mechanically, or to have reacted automatically, without thinking, and so on.

Obviously, our accounts of action and behaviour are constrained by assumptions about the acting entity. But this formulation can be misleading if it is taken to suggest that 'assumptions' pre-exist, or are antecedent to, descriptions of action and behaviour. For there is a reflexive tie between the practical description of action and the 'assumptions' which this description exhibits. Our discourse on entities thus both presupposes and exemplifies assumptions about the nature of the entity.

### *II.2 The object hypothesis*

There is a yet further level of complication which is worth stressing for the purposes of this paper. In the discussion so far, we have assumed that entities are relatively distinct, discrete and discernible. We have operated, in other words, on the assumption that entities are *bounded*; that they can be differentiated from other entities and from the environment in which they are situated. Much of our mundane discourse thus presupposes and exemplifies the assumption that the characteristics of an entity can be associated with a particular bounded space. This is part of what Wright calls the object hypothesis (Wright 1990a). He suggests that one of the biggest questions which philosophers beg is whether there are objects or whether there is a continuum, a 'flux of varying viscosity' (Wright 1990b). For Wright, ontological gerrymandering (Woolgar and Pawluch 1985) begins with the belief that there are already delimited objects in what exists. What is not sufficiently recognised is that the object hypothesis is a convenient fiction, a method presumed necessary for ordering (representing, describing, accounting) the world.

Are there objects or is there a continuum? We do not need to

tackle this philosophical question head on in order to recognise the benefits for our analysis of assuming the counter-intuitive answer. By taking the flux of varying viscosity as heuristic, we see the world as comprising a range of objects whose boundaries exhibit varying degrees of definition (in both senses of the word). This view throws into relief one of the foundational ordering principles of our phenomenal world: the presumption that entities are bounded.

It is fairly easy to show that conceptions of boundedness are both culturally and historically relative. We do not need to travel far into the anthropological literature to find instances which confound our modernist prejudices. For the Irku of the Eastern Sahara, for example, the centre of a person's consciousness (*frut-rot*) is said to reside in the nearest Carob tree, that is, outside of – but spiritually tied to – the human bodily form (Menteur 1928). In corroboration of this view (and in a startling resonance with the modern Western experience of 'jet-lag'), certain forms of rapid travel are proscribed, lest the individuals experience severe disorientation brought about by disjunction from their centre of consciousness. By contrast, in Western Namibia, the Prurati do locate the 'soul' within the body, but at an unusual specific location: slightly below the left elbow (Phillpott 1974). This accounts for – and is reaffirmed through – a whole series of otherwise inexplicable gesticulations which feature in the dances at the centre of healing rituals.

Such examples both underscore the problem of assigning characteristics to entities, and challenge our assumptions that entities are discrete in the first place. Clifford's recent (1990) re-evaluation of Menteur's classic work suggests that his Irku informants might not have been nearly as clear-cut in their assessment of body-consciousness relations as Menteur's final report indicates. Instead, we may be reading the result of Menteur forcing his Western predilection for the boundedness of entities (the body separate from the consciousness) upon the puzzled efforts of the Irku to keep their French visitor happy.

Less exotically, of course, we recognise changes over time in the attribution of characteristics and capabilities to entities within our own societies. For example, in conservative philosophies, responsibility for deviant behaviour resides with the deviant. The origin of the behaviour is essentially located within the deviant herself and, as a result, remedies emphasise the virtues of removing such deviants from society. By contrast, more liberal philosophies

locate the origin of deviant behaviour outside the deviant, and largely beyond the deviant's control. The focus of responsibility shifts to various circumstances antecedent to the deviant herself: her upbringing, family background, peer relationships. From this vantage point, the deviant's characteristics do not inhere but are attributed, assigned as part of a social process. As a consequence, remedies are directed more to repairing the adverse effects of antecedent circumstances (through counselling, treatment) than to removing a presumably unchangeable deviant. Similar examples are found in the social problems literature, where differences between assignation and inherence views also emerge in debates about various non-humans and inanimate objects: for example, drugs and other substances, coffee, margarine, alcohol and so on.

We see that we can question both our assumptions about the characteristics of various entities and our assumption that entities are discrete, bounded things in the first place. In this postmodern world, then, nothing about entities is fixed. Worse, nothing about them is certain. That they are a thing, that they constitute an entity is as problematic as assigning certain capabilities and characteristics to these things.

This way of putting it replays a key problem in social theory. Do characteristics reside in, or are they attributed to entities? As we have suggested, our answer to this question has important implications for the adequacy of explanation and for issues of responsibility. For example, the difference between natural and social science, the Winchean view that social subjects must not be treated as natural objects, is premised upon profound assumptions about the difference between social and natural objects. More classically, this same issue arises in the debate between free will and determinism. Are human actions to be understood primarily as the exercise of free will or as the result of forces outside of, and beyond the control of, individuals?

### *II.3 The moral order of representation*

As we have suggested, answers to this kind of question vary enormously, both over time and cross-culturally; according to which kind of entity one is discussing; and according to which notions of boundedness inform the discussion. Importantly, views about the capability and boundedness of any one entity are bound up with views about other entities and of the relationship between

them. For convenience, we can refer to any existing complex of relationships between entities as the moral order of representation (Woolgar 1989). It is a world view which embodies notions about the character and capacity of different entities, the relationship between them, their relative boundedness, and the associated patterns of rights and responsibilities. Linking all these are sanctioned procedures for representation. Thus, one knows and can adequately represent the views of any one entity in virtue of how it relates to and differs from others in the matrix.

The moral order of representation changes with the introduction of a new entity. At a micro level, for instance, the introduction of a new item of apparatus in the scientific laboratory might have important consequences for the rights and responsibilities of existing pieces of apparatus. At a more macro level, the emergence of machines dubbed artificially intelligent can revise assumptions about (natural) intelligence in non machine entities. Or, ideas about what distinguishes humans from animals can change as the result of episodes such as the ape-language controversy. The research reported in this paper looks in detail at a less cosmic alteration to the moral order: the design and introduction of a new range of personal computers. Note here that the focus is not the advent of a new kind of technology, but rather the introduction of a new variation on existing technology.

## III An ethnography of computers

In order to maintain ambivalence about the appropriate unit of analysis, as a way of exploring assignations of agency and changes in the moral order, the study reported here is designated an ethnography of computers. I carried out an eighteen-month participant observation study in a medium-sized company which manufactures microcomputers and allied products, primarily for education. (Since certain members of the company are still uncertain about the benefits of publicity arising from the kind of analysis undertaken in this paper, I currently refer to the company anonymously.)[8] They are a phenomenally successful company, having been founded some 14 years previously. The company had grown in size by an average of approximately 20 per cent per year over the last 5 years, and its turnover had increased by an average of about 35 per cent per year in the same period. By the time of my study they had achieved a position such that both they and their

main competitors were claiming in excess of 50 per cent of the market share.[9]

My original research design was to follow a major project in detail from inception through to launch, first shipment and after sales feedback. After some negotiation, it was agreed that I should join the company as part of the newly expanded project management team. We felt this would be a strategic position from which to carry out the study since, as a project manager assistant with responsibility for liaison and co-operation between different sections within the company, I would be able to enjoy relatively free access across disparate parts of the company. In particular, I worked as a project manager assistant on the project designated 'DNS'.[10] In broad outline, the aim of the project was to produce a new range of microcomputers built around the new 286 chip. As fairly soon became apparent, this entailed following the lead established by IBM in the production of their IBM PS/2 standard.

The DNS range was the third in a recent series of microcomputer product ranges which brought the company more into line with IBM compatibility standards. The first of these – the 'Stratus PC' – had been built around the 186 chip in order to 'provide an educational computer which was appropriate for schools'. I was told that the Marketing section had received the acclaim of the press for the Stratus PC with some glee, especially when one review went so far as to praise the machine by speaking of the IBM PC as a good Stratus clone. In fact, the Stratus PC was not designed as IBM compatible, and although IBM was not at that point seen as the main competitor, a further range – the 'K series' – was developed to compete with the IBM XT at the high end of the market. Subsequently, DNS (subsequently marketed under the name Stratus 286) was developed to fill a position between the two previous ranges, combining the educational virtues of the Stratus PC (186) with the IBM compatibility of the K series (286).

## IV Configuring the user

The earlier discussion of attribution and boundedness in entities suggests we start from the position that the machine (in this case, DNS) can only be understood in terms of its relationship with other entities of its phenomenal world. However, this recommendation is not simply a call for understanding technology 'in its context', since the nature of 'the context' is itself subject to all we

have said about the nature of the machine (cf. Cooper 1990). The character of both entities is essentially indefinite; and the character of both entities is reflexively tied (Garfinkel 1967), In other words, representations (descriptions, determinations of many kinds) of 'what the machine is' take their sense from descriptions of 'the machine's context'; at the same time, an understanding of 'the context' derives from a sense of the machine in its context. The sense of context and machine mutually elaborate each other. For that aspect of context called the user, the reflexive tie is especially marked. The capacity and boundedness of the machine take their sense and meaning from the capacity and boundedness of the user.[11]

Less obscurely, perhaps, our textual metaphor makes the same point. Construing the machine as a text encourages us to see that the nature of an artefact is its reading. But in trying to escape the dreaded technological determinism, in disassociating the upshot of reading and interpretation from any notion of the inherent quality of the text (what it actually says, what it actually means), we do not mean to suggest that any reading is possible (let alone that all readings are equally possible). In principle, this *is* the case. For example the dictates of sceptical ethnomethodology (or of ethnomethodology at the hands of some interpreters) pose an idealised User/Reader, one unfettered by relationships with other texts.[12]

If, however, we wish to acknowledge that in practice only a limited set of readings is possible, our question is how to account for this delimitation. Following Smith (1978), we can suggest that the organisation of the text makes one or other reading differentially possible.[13] For Smith, the important point is that the organisation of the text is isomorphic with the concept we use to make sense of it. In other words, for example, a text 'about' mental illness will be organised in such a way as to make this reading possible. By direct analogy, I suggest, the machine text is organised in such a way that 'its purpose' is available as a reading to the user. In her analysis, Smith notes how certain organisational features of texts provide 'instructions' which enable readers to make sense of content in terms of conclusions stated at the outset. To adapt Smith's terminology to our concern with technology, the user is encouraged to find in her dealings with the machine an adequate puzzle for the solution which the machine offers.

A small extension of this analytic stance on texts suggests that the organisation of the text hinges not so much on mundane

features like the length of sentences, the amount of space devoted to different topics etc, but rather on associations made available within the text and between text and reader. Textual organisation refers critically, as far as the sense to be made of it is concerned, to the relationships made possible between the entities within and beyond the text. Certain characters become central to the story and others peripheral; groups of actants join forces while others disperse; the activities and achievements of some are highlighted, while others are relegated to the background, silenced and unnoticed. The reader (who is, I am afraid, the writer) of the text is invited to join with certain groups and disassociate herself from others. A simple example is the invocation of community through the use of the royal 'we'. (Of course, this is an example with which *we* are all familiar. Do you, gentle reader wish to say you are not familiar, and hence risk being excluded from our text?) The text might be said to be designed (perhaps implicitly, perhaps unconsciously, but always within a context of conventional resources and expectations) for the reader. What sense will she make of this (or that) passage?

In configuring the user, the architects of DNS, its hardware engineers, product engineers, project managers, salespersons, technical support, purchasing, finance and control, legal personnel and the rest are both contributing to a definition of the reader of their text and establishing parameters for readers' actions. Indeed, the whole history of the DNS project can be construed as a struggle to configure (that is, to define, enable and constrain) the user. These different groups and individuals at different times offered varying accounts of 'what the user is like'. Knowledge and expertise about the user was distributed within the company in a loosely structured manner, with certain groups claiming more expertise than others in knowing what users are like.

## V The distribution of knowledge about users

### *V.1 Difficulties of knowing the user from within the company*

My first vivid introduction to the socially structured character of knowledge about users occurred during an early meeting of a group of technical writers in the User Products section. The discussion centred on plans for carrying out usability trials. Who should be invited to act as subjects for these trials? The problem

was that constraints of confidentiality made it difficult to select subjects who would know nothing about the new machine. Getting in 'the man on the street' as they put it, was not a realistic option. At this point, parties to the discussion started to consider the possibility of finding 'true novices' amongst the work force in the company.

In a fit of helpfulness, I offered my services. If they were really stuck, I said, I would be very happy to act as a subject since I would have no problem in acting as a naive user. I was very surprised when SallyP turned on me vehemently. Didn't I realise how differently users approached this? Didn't I realise how extraordinarily difficult it was for anyone in the company to appreciate the way users looked at things? People in the company couldn't possibly appreciate the user's point of view. Indeed, this was a major problem which pervaded the company: a failure to understand what it was really like to be a user.

I realised I had inadvertently stepped out of role. As a relative newcomer to the company, I had expressly volunteered my naivety in the Company Perspective. But SallyP was apparently unaware of my 'real' identity as participant observer. She assumed I was part of the project management team, and it was in that guise that she was addressing me. I had unwittingly reaffirmed her worst fears about insensitive 'techies' and their inability to see beyond a company mind-set. So I was admonished for presuming to be able to act like an outsider.

### *V.2 Alleged deficiencies in company knowledge about users*

As this last anecdote suggests, certain individuals could claim the right to speak authoritatively on behalf of users. At the same time, it was said that some of the individuals and groups you would expect to know about users were manifestly deficient in just this kind of knowledge. For example, one of the technical writers spoke of her amazement in discovering the attitude towards users in Marketing:

> You can find the same thing at Marketing. I remember going along and saying 'Excuse me but can you tell me who the target market is for this?', you know. And they'd *looked* at me, sort of thing. Well I always thought Marketing had, you know, like a list of and a target market would be durhrhrhrhrhr education, a

> sort of list of generalisations. No way! Nothing like that! So there's no guidance like that. (user01:CH)

I was told by a long serving member of Technical Support that 'typically, the engineers don't have a clue about users'. She told the tale of an early attempt by 'engineering' to encourage users initially to configure their new machines by input-ing a long line of characters which would have been 'meaningless to your average teacher' (user09:RC). She poured scorn on what she saw as the engineers' presumption that users would be happy to have to do this.

### *V.3 Stories about users*

Members of the User Products section felt that their conception of users was affected by a wide range of influences, ranging from their own first time of using computers through to 'hearing tales about what happens outside' (user01:CH). Knowledge about users thus involved the circulation of stories and tales about the experiences of users. Frequently, stories about 'what happens outside' seem to have originated in the Technical Support and Service sections of the company.[14] These sections were generally reckoned to represent 'the sharp end' of dealings with customers. The view was that whereas, for example, the engineering and design sections worked in some isolation from users, those in Technical Support had much more experience of users since they dealt with user complaints and queries at first hand. Some of those in Tech Support had themselves 'moved out' from working in the engineering sections and seemed keen to emphasise their new found responsibilities in dealing with users.

Significantly, these stories about users were nearly always couched in terms of insider–outsider contrasts: what was happening (or had happened) 'on the outside' was a recurrent motif. The boundaries of the company thus played an important part in the telling of these tales. For example, one respondent recalled the experience of one particular 'outside' visit in the following way:

> Some of us have been out to visit users but it was something that was thought of as a good idea but never really took off. . . . I went out once something like back in '84, a long time ago and it was actually a (roadshow) because I went up to a school and I

> just remember seeing this room of computers, a square room and they were in sort of an L shape against the wall. And pinned above them were very very simple instructions for what to do. And they weren't, they looked like they'd been used, you know? It was almost as if they'd been unpinned, taken down and pinned up again and again so that someone had had them right by the machine.
> (user01:DI)

This experience of an 'outside' visit thus led to the realisation that in schools someone, perhaps a teacher, had had to devise extremely plain instructions for use alongside the machine. The same respondent related a story passed on to her by a colleague in Technical Support:

> Another tale I remember hearing is that a school who had a machine up to like four months. They wouldn't unpack it or anything, they were too scared. There was no one around they thought was able to do much with it. . . . Yes. I mean *GOOD GRIEF!*

### *V.4 User singular and users multiple*

Whereas participants often referred to 'the user' in the singular, it is not clear they thought users of the Stratus 286 would all exhibit identical, monolithic sets of attributes. They could presumably imagine a wide variety of purposes and uses; they would have been aware that Marketing stressed the versatility of the machines when promoting the company's products. Clearly, one criterion for a successful text is precisely its appeal to a wide range of users.

> There's a limit to how far you can take what any user or set of users wants into account when you're designing a product. It would have been very easy for us to say we want this product to be suitable for teachers in secondary schools, what they want to get out of the machine. We could have produced a very watertight specification of what the thing had to do. But what we knew was we wanted to cover primary schools, secondary schools, colleges, universities, business users, government users, CAD people. The trick was not in finding out what one set of users wants, because if you limit it to a small enough

> number that's fairly easy, the trick was trying to find that area of overlap that would suit them all, get the best fit. What do you mean by best fit? Who knows?!
> (TL 9-11-90)

The text sells well if many different readers find a use for it. One might even go so far as to say that an author's attempts to prescribe readings, to delimit ways in which the text can be read, is a sure recipe for disaster, at least in the sense of guaranteeing early returns from the publisher (haha!). So a strict and narrow definition of user would seem counter-productive. Similarly, user configuration which restricts the range of possible readings will not generate success.

All this makes curious the continued singular reference to 'the user' in the company, until this is understood as a generalised formulation produced for purposes of establishing contrasts between insiders and outsiders. The generalised user provides a more successfully stark contrast with us/the company/members of the company than would a heterogeneous rag-bag of customers with varying attributes. The contrast is rhetorically important for example, as we have seen, in stressing the difficulties of knowing what precisely it is that users want. Given the extent of the (claimed) differences between the way 'we' look at the world and the way 'the user' looks at the world, it becomes necessary to rely upon especially skilled spokespersons – those few with knowledge of these very different entities. When someone in User Products says that Engineering have no notion what the user expects, the achieved distinction between the monolithic entity – the user – and the monolithic entity – the engineer – makes a political point about the inadequacies of all members of Engineering. More pervasively, this generalised formulation reaffirms divisions between us and them. Company boundaries, differences between insiders and outsiders, are more greatly emphasised through deployment of 'the user' than by admitting that some users are more familiar with our machines than others. This rhetorical rendering of the generalised user also afforded some interesting variations on the more familiar examples of prejudiced rhetoric: He was a user but he seemed to know what he was talking about.

*V.5 Users don't necessarily know best*

References to the user emerging from the Engineering sections of the company included the view that, although it was important to have an idea of who 'the user' was and what they wanted in the machine, users' views should not be unproblematically adopted in design.

> The user isn't necessarily able to see in a clear enough way each feature at a price that they're prepared to pay for it. I don't believe you can go to a user and say right each of these functions we're thinking of putting in the machine, tell me how much you are prepared to pay for each of these. I don't think you can construct a product specification like that.
> (TL 9-11-90)

The suggestion was that design should respond instead to ideas about 'where the market was going' or 'where things were going', a more generalised conception of the future requirements of computing. Significantly, such conceptions were frequently referred to as 'visions' of the future, which seemed to stress technical progression and which were couched in terms which transcended individual users' desires for particular technical features.

> Where the clever bit comes in is people like [the Managing Director] having a vision and saying we're going to do this and being able accurately to predict that if we don't do that we'll still sell the required volumes of the product without delaying it. Or without putting in this feature which [the users] might have said was desirable but which they didn't really want to pay the cost of.
> (TL 9-11-90)

A variant of this line of argument was the more familiar view that there was no point in asking users what they wanted because they themselves didn't know. According to this view, such ignorance arose primarily because users were unaware of likely future developments:

> Users can only know about what's available at the moment. So they'll tend to give you an answer that's based on different

> combinations of what's available at the moment. What we're trying to do is to make available to them something that isn't available at the moment. Which is where the [Managing Director] visionary idea comes in: We ought to be doing this because I say so and because I know what I'm talking about! (TL 9-11-90)

We see here an effective rationale for not placing too much emphasis on users' views. According to this perspective, configuring the user involves the determination of likely *future* requirements and actions of users. Since the company tends to have better access to the future than users, it is the company's view which defines users' future requirements.

## VI The usability trials

### *VI.1 Background*

The usability trials are, of course, just one occasion where articulations of 'what the user is' featured prominently. As we have already suggested, myriad other events and occasions during the course of the project included fairly explicit attention to the question of the character of the user.[15] More generally, of course, determinations of the user could be seen taking place throughout the construction of the machine-text. It is thus possible to argue that participants' notions of the user are available to us, if only implicitly, through an inspection of, say, the day to day work of the hardware designer. The interest of the trials, however, is that they involve explicit articulation of whether or not prevailing ideas about the user are correct. The matter is made explicit, in the case of the trials, through an assessment of the different courses of action which a user might engage in.

We have already mentioned that the company encompassed a variety of perspectives on the importance of taking users' views into account. This makes it difficult to be clear to what extent the upshot of these particular trials had any consequential effect on 'settling' the question about the nature of the user. Although there was, as we shall see, some concession to experimental method in the design of the trials, the results were never written up in a final form, to be circulated to designers and other members of the project team. Instead the 'results' tended to be fed back piecemeal

into the production process. For example, when one of the test subjects had trouble understanding a diagram on page 34 of the Stratus Guide, this information was quickly passed by word of mouth to one of the technical writers, who then redrew the diagram for the next draft. So it is difficult to discern any clear outcome of the trials which might stand as a definite milestone in the ongoing configuration of the user. Nonetheless, these trials were thought important, at least by members of the User Products section of the company. This section devoted approximately 6 person-weeks to carrying out the trials; it would have been more but for the delays and time pressures already mentioned above.

### *VI.2 The importance of the case*

The start of the trials was delayed several times. The User Products section was caught, as it seemed to be on several other occasions, between the need to ensure usability testing as early as possible in the development of the product and the late availability of a 'finished' product. It was reasoned that the most fruitful assessments of usability could only be carried out with the product in a form as near as possible to that which would be experienced by the user. One of the main reasons for the delay in the project as a whole centred on the availability of the case. Members of User Products took the view that usability trials could only properly take place when a cased version of the machine was available. Some negotiation ensued when the first prototype case appeared, but Product Engineering argued that it was too risky to loan the sole case for purposes of usability testing.

It is significant that User Products felt the necessity for a physically bounded entity for use in usability testing. The machine would not be a real machine unless it was in its case. 'Real' in this usage specifically denotes 'the kind of machine a user would expect'. This contrasts markedly with what counts as a real machine within the company. Particularly within the engineering sections (notably Hardware Design and Engineering Quality), machines were mostly left open on desktops and workbenches, their innards displayed, precisely so that the engineers had quick access to the inside of the machine. In these sections, it was unusual to find a machine inside its case.

The following contrast between the treatment of computers

'outside' and 'inside' the company was provided by one of my students employed by the company for a summer work placement:

> When I joined the company I was a 'soft' user (Turkle 1984). Although I did not believe the computer was 'magical', I could not recognise the internal parts of a computer and had never taken the casing off a computer. In fact I had always been deterred from doing so. However in the EQ [Engineering Quality] section, no such squeamishness was expected. Machines were perched on 'breadboards' – metal frames or boards, or they were missing their top covers. . . . At school I had been told that the ideal place for computers was a dust free atmosphere kept at a controlled temperature. In the company, there was no such reverence for the computer. They were regularly taken apart. In fact, when a machine which was in its case did not work, the top was removed immediately and the boards were jiggled around just to check that the connections were all right. (Dobbins 1990)

The surprise of finding the innards of computers regularly on display around the desks and benches in the company is part of the experience of moving from the outside to within the organisation. The machine's boundary symbolises that of the company, so that access to the inner workings of the machine is access to the inner workings of the company.[16]

The symbolic importance of the machine case/company boundary also features in the 'induction programme' – a series of meetings and events arranged over a period of two or three weeks for those starting with the company. I visited or had meetings in product engineering, hardware design, purchasing, personnel, marketing, engineering quality and so on. But the generally acknowledged highlight of the programme was the visit to manufacturing. (When they learned I was undergoing 'induction', a first question from friendly colleagues was whether or not I had 'been down to Manufacturing yet'). This meant spending an hour on the assembly line under the tutelage of Rose. Rose did all the manufacturing inductions. She explained the sequence of operations for building a Stratus (at the time of my induction, the Stratus PC) and then asked me to try my hand. I could not believe I was to be entrusted with putting one of these things together! Like my student and most others new to this experience, I was amazed that mere novices were encouraged to handle the very insides of such a

revered item of technology. Rose guided my nervous efforts with a matter-of-fact patter borne of long experience with similarly incredulous newcomers. 'Just turn over the frame now. This way. That's it. Have you got your board. Right. Put your first screw in there. That's it. See, it's not so difficult . . .'. Although ostensibly just one of a series of events designed to familiarise the new employees with different parts of the company, this 'hands-on' experience can be understood as a symbolic welcome into the company (machine) by way of dis-abusing computer primitives (like me) of the mysteries of computers.[17] As a result of this experience, I remember thinking that the 'real' sophistication of the machine must lie elsewhere, perhaps in the printed circuit boards. I felt I had penetrated the outer shell of the company, but not yet its heart, the nitty gritty of technical design (hardware engineering).

During the later stages of participant observation, the possibility arose of my buying one of the new Stratuses. As a bona fide member of the company – for such I would construe myself for these purposes – I would be entitled to buy one at discount. But in deciding whether or not to do this I was struck by the way my assessment of the machine changed according to my (and its) location. On days away from the company, I had a good sense of what the machine could do, was for, looked like. I had a confidence in it. (It is, after all, a very nice machine.) These feelings were not unconnected with the fact that I was its representative on the outside. I could talk authoritatively to my Brunel colleagues about this new machine; I had privileged information about it. Clearly, on these occasions the machine I knew about was 'Marketing's Machine'. It is with some embarrassment I now recall my conversations with BobT, the sales director for higher education, about my sounding out the market for the company's products at Brunel. There was even an occasion when I handed out Stratus 286 brochures as part of a talk at Brunel. It is, after all, a very nice machine.

By contrast, on days in the company, I often found it difficult to imagine how the thing could ever work (cf. Collins 1986; Mackenzie *et al.* 1988: 161–2). The case was delayed again, the toolmakers had been taken into receivership, the chip suppliers had welched on their delivery dates yet again, the winchester access times were way down on target, MartinK had been taken off the project because of problems with 186 deliveries, and so on. When TedJ, a senior member of the hardware team, told me it

would be wise to wait at least 6 months after launch before buying my own Stratus 286, I could see his point. He was sharing with me his view from/on the inside of the machine. Insiders knew that although the initial machines would *look* okay, a great deal of patching up had gone into them for purposes of just 'getting them out of the door'.[18]

These and similar examples underscore the symbolic importance of the machine's (text's) boundary. The video record of the usability trials shows putative users working out how to relate to (and in one instance, literally, how to *connect* to) a technology which has already been black-boxed. Or, in this instance, beige-boxed. The task for subjects of the usability trials is to work out how to access the interior of the beige box, in order to extract what they need from the machine/company. The machine's task is to make sure these putative users access the company in the prescribed fashion: by way of preferred (hardware) connections or through a predetermined sequence of keyboard operations. The user will find other routes barred and warnings posted on the case itself. Labels bear warnings of the dire consequences of unauthorised boundary trangression: electrocution, invalidation of the warranty and worse:

WARNING
LIVE PARTS ARE
EXPOSED IF COVER
IS REMOVED

Guarantee of safety and product warranty void if seal is broken.

Inside the case (assuming we allow ourselves access for a moment), users find that different modular components of the PC are similarly labelled, thus structuring and guiding access within and around the machine (company). In particular, various makes of disk drive bear a variety of warnings:

Warranty void if cover is removed or this seal is broken
(IBM 30Mb Winchester)

Warranty void if this seal is broken
(IBM 60Mb Winchester)

The 40Mb Seagate drive bore three labels:

Product warranty will be Void if this label is removed

Do not apply pressure to top cover

Delicate Equipment
HANDLE WITH CARE
Disk/Head damage may occur

For those modular products supplied by the company as replacements or upgrades to the machine, warnings posted on the product were sometimes accompanied by injunctions to contact the company in case of doubt. For example, the following appeared in black capital letters on a glossy yellow sticky label, affixed to a replacement hard disk drive:

WARNING:
*STATIC SENSITIVE DEVICE*
FAILURE TO OBSERVE THE FOLLOWING WILL INVALIDATE YOUR WARRANTY
★ *DO NOT DISCONNECT* THIS HARD DISK DRIVE WITHOUT USING A WRIST BAND
★ *NEVER* DISCONNECT THE CABLE FROM THE DRIVE
★ *NEVER* TAKE THE GOLD FINGERS OFF THE DRIVE OR CABLE

IF IN DOUBT CONSULT YOUR USER DOCUMENTATION OR TELEPHONE [THE COMPANY] TECHNICAL SUPPORT HOTLINE ON 0898-239239

Here we see that, in the event of uncertainty, users are redirected back to sources – either 'user documentation' or the company technical support hotline – which can re-establish the correct pattern of user action, in line with the approved configuration of the user's relationship with the company.

### *VI.3 The manuals*

Ostensibly, a central concern of the usability trials in which I participated was to evaluate the draft documentation which was to accompany the machine on its shipment. The main body of documentation comprises the Setting Up Card, the Stratus (286/386) Guide, the Reference Diskette, the MSDOS4 Users' Guide

and the WINDOWS Guide. The first three of these items are produced by the company and relate specifically to the operation of the Stratus 286. The latter two relate to bought-in proprietary products which are supplied with the machines. In addition, peripheral equipment supplied with the machine, such as a printer, came with further documentation specific to its own use. The company specific documentation was a main focus of the trials, but participants were also keen to evaluate the relationship between the other items of documentation. Would users be able to select the correct item of documentation when attempting to solve a particular problem? Were the instructions in, say, the Stratus Guide, sufficiently clear in telling users which other parts of the documentation to consult and when?

The body of documentation at the centre of the trials comprises a set of texts which accompany the machine which, we suggested from the outset, is itself best understood as a text. We can think of the documentation texts as peripheral texts intended to enable the operation/reading of a core text. They are, so to speak, captions for helping readers find and see the relevant features of the machine itself. These captions configure the user in the sense, discussed above, of defining the correct courses of interpretation and action to be followed. They help guide access to the machine text. Long sections of the video records of the usability trials show subjects moving back and forth between manual-text and machine-text, seeking the sense of a described feature of the machine in the material object itself, and assessing the sense of one of the manual's instructions in the response of the screen to some keyboard operation.[19]

A central concern for testers/participants was whether these peripheral texts were sufficiently 'clear' to users. They were sometimes said to be 'clear' if subjects were judged to have understood and/or carried out the tasks set them by the testers. The manual-text can thus be seen as having enabled operation of the machine-text. As we shall suggest, determinations of the relative reliability of different texts were managed by construing a *distance* between them, such that one was viewed as operating 'at a different level' from another.

We have already suggested that the trials include detailed articulations of 'what the user is like'. However, it was not enough to determine whether or not a subject could fulfil a task. The testers were also interested in knowing whether the subject had carried out the task in the manner a user would have done. The

trials can thus be understood as an occasion where the machine and its documentation confront (a version of its) user(s).[20]

What is especially interesting is that, at this stage in the project, the identity and capacity of the key entities involved is equivocal. This means, on the one hand, that the capacity of the machine, what it can do, what it is, whether or not it works and so on, is not yet settled. By this, I mean to claim that the trials show participants' awareness of the possibility that the machine is not (yet) working as required, that things might yet go wrong. In this usage, 'settled' refers to participants' projection of particular states of readiness of the machine e.g. ready for launch, ready for shipment and so on.[21] Similarly, at this stage, the identity of The User is not settled. Although participants could and did trade versions of what users are like, the identity of the user of the DNS/ Stratus remained essentially uncertain.[22]

This makes these trials interesting in respect of accounting for interaction between machine and user. Where IT novices use established IT products, a typical experience is that where things 'go wrong' the 'fault' is likely to lie with the user. Conversely, where experienced users of IT products come into contact with machines still under development, the fault can be more readily said to lie with the machine.[23] Of course, the determination of things going wrong does not rest solely with the human agent. The machine may declare 'error' as a way of indicating that the user is at fault ('Printer Needs Attention'), or the machine may self-diagnose error ('WP.SYS file not accessed').[24]

In the DNS/Stratus usability trials, neither machine nor user is settled/experienced/established. Consequently, the interactions are part of the process of establishing the identity of the interacting entities. In other words, in this situation, the interaction between machine and user invites assessment both of whether or not the machine is acting like a real machine *and* whether or not the user is acting like a real user.

### *VI.4 Getting the right context*

In planning the trials, particular attention was given to the selection of subjects and to choosing the right locale.

How can we find subjects who are most likely to act like users? A standard procedure for manufacturers, especially in electronics and IT industries, is to use what are called 'beta sites' – trusted and

privileged customers who are happy to try out new products. These customers gain advance notice of the release of the new product in exchange for feeding back information about how the product can be finally improved. But the company had little or no tradition in the use of beta sites.[25] In any case, a main aim of the trials was to try out the machine (and its documentation) on relatively novice users. Trusted customers with a close relationship with the company were unlikely to fit this particular requirement. It was suggested that a group of students be recruited from the local polytechnic. This had the practical advantage of being easily arranged through a local contact; in addition, such a group obviously matched one of the main customer target sectors (further education). However, this idea was rejected because it wasn't thought possible to maintain the necessary level of secrecy.

The need for secrecy, in particular, forced the User Products group to consider selecting people from the company. The problem about secrecy was thus effectively finessed, but the level of these subjects' expertise still remained a problem. As a result of drawing upon personal contacts in the company, the following individuals eventually served as subjects: two members of the night shift from manufacturing; the head of User Products; a psychologist from the local polytechnic (who was working temporarily with the company); a technical writer; and a project manager assistant (me!).

Where should we carry out the trials? As in most experimental situations, the answer hinged on a compromise between a setting which best approximated the subject's 'natural' environment and a setting which facilitated the kinds of observation thought necessary for the conduct of the trials. Some larger companies have a small closed-off office space for this kind of testing, which they designate 'the laboratory'. But at the company where I carried out this research, space (especially closed-off space) is at a premium and usability trials are not thought sufficiently important to warrant a specially assigned area. The trials were held in the main sitting room of a Victorian terrace house in Race Street, a few hundred yards from the main factory site. This house, recently modernised and refurbished, was being let to the company as temporary lodgings for newly recruited middle management.

The video record shows several features of what might be regarded as a typical user's environment, present in the Race Street setting. In addition to the test subject (the user), the Stratus 286 and its accompanying peripherals and documentation, there

were table, cups of coffee, chairs, television, bookshelves, carpets and so on – all the various accoutrements of being in an ordinary place. Indeed, the video record suggests that coffee drinking featured prominently in the re-creation of the users' environment. The telephone line which enabled instant phone calls to the company proved an especially useful feature of this users' environment. The kinds of feature presumably *not* present in the typical natural user's environment included: the testers (observers) with their clip boards, notepads and clocks; the video camera; the (audio) tape recorder;[26] and other machines (such as the more powerful K series computer). When viewed from a non-specist perspective, this is a comical concatenation of entities.

## VII Analysis

### *VII.1 The form of the trials*

Each trial started along similar lines, as indicated by the transcript (of the audio recording) in Appendix 1. The subject was confronted with machine, peripherals and documentation. The tester explained the general purpose of the trial, pointed out the equipment available, set a task and asked the subject to say how s/he might go about it and to estimate the length of time it would take. The bulk of the trial comprised the subject then trying to complete the task. Finally, there was a 'post mortem' when tester and other observers would discuss the trial with the subject. The whole event was both video and audio taped.[27] In the particular interaction transcribed as Appendix 1, the video record shows some initial confusion as the various human participants attempt to get into position before the start of the trial. The observers bump into each other as they move around the table. They strive to achieve what they regard as their appropriate juxtaposition vis-à-vis the machine for the purposes of the conduct of the trial.

### *VII.2 Constructing natural users*

The central part of the trials is particularly interesting. The testers cast themselves as objective observers in the sense of not wanting to intrude upon the 'natural' process of a user trying to make sense of the situation. They want an unbiassed picture of how users

'actually' go about the completion of the tasks. On the other hand, a whole series of (thoroughly .unnatural!) contingencies arose which demanded their frequent intervention. For example, where subjects were thought to be going hopelessly wrong, or where they were clearly about to get into trouble, it was felt necessary to retrieve the situation.

Quite apart from this kind of intervention, however, observers offered considerable commentary on subjects' performance:

1. (P/A/S 28:30)
   A: You actually succeeded in this task, so there's no problem about that

2. (P/M/N/A 1:26:18)
   P: You're a technical author's dream – reading the manuals!

In a situation where the identity/capacity of both machine and user are unsettled, we might expect participants to express concern over exactly who or what was being tested. The recurrent commentary on the subjects' performances – which is presumably not a 'natural' feature of the user's environment – can thus be understood as the observer's effort at reassurance about the real subject of the test. Frequently some confusion – over who (or what) was carrying out the task – revealed itself in the observer's attempts to empathise with the subject:

3. (SP/A/S 30:30)
   A: Let's assume we succeeded there which I think you did

Observers frequently intervened to explain the origin of a problem in terms of a machine fault, where this prevented (or made difficult) the completion of the task by the subject:

4. (SP/A/S 29:05)
   A: It's a hardware error (3.0) probably a loose connection (3.5) you always have these problems on pre-production. But why did it have to happen in the middle of a trial!

5. (R/N/P/S 2:22:14)
   R: I'm so pleased it wasn't me this time huh huh.
   N: You've done fine so far Ruth

A large number of prompts and interventions seemed to pursue the issue of whether or not the subject was acting sufficiently like a real user. In each case, the tester explored with the subject the way they would behave, if they were in fact acting like a real user:

6. (SP/A/S 34:01)
   A: You'd know WINDOWS was on there
   (2.0)
   A: I *think* you'd know that wouldn't you?
   SP: Yeah
   A: That's one reason you'd buy it!
   SP: Hmmm yeah yeah

7. (P/M/N/A 1:37:43)
   N: Just do it as if you were doing it normally

8. (R/N/P/S 2:26:03)
   N: This wouldn't normally happen with someone who's been doing something with it already

9. (SP/A/S 48:57)
   A: Of course you would know how to use WRITE
   SP: I've used WRITE before so it would take me longer

But the participants were not above ironicising their own attempts at creating an objective test of 'natural' user behaviour:

10. (SP/A/S/ 53:01)
    A: Do you want a rest now
    SP: Yeah
    A: A coffee?
    SP: Is that one of my tasks? 'Make the coffee and tea. How long do you think it would take you' Hah hahn.
    SW: 'Subject drank thirteen cups of tea!' Huhuhuh

### *VII.3 The 'wrong socket' episode*

An especially vivid illustration of many of the themes already discussed occurred in one particular trial, when Ruth was asked to connect the (new) Stratus 286 to a printer. In order to see if the 'machine' worked – and by 'machine' we can here understand

the configured relationship between Ruth and the Stratus – the observers used as criterion the successful operation of a 'peripheral'. This reflects the fact that satisfactory usage of a machine often requires users to invest in and deploy auxiliary items of apparatus. For this reason, it is worth considering this part of the interaction in detail.

[Reader: play video tape Extract B here]

[[For any readers without this video tape extract:

Copies of the video tape extracts are available as an optional extra to this paper, price £14.95. Order from CRICT (address on the cover of this paper) quoting 'R/N/P/SW 2:17:15–'.]][28]

The audio record of this interaction is transcribed in Appendix 2. At the time we join the scene, Ruth is confronted by the Stratus 286 (with its keyboard and monitor); various instruction booklets and an as yet unconnected printer. The Stratus is initially switched on.

Ruth begins by asking the observers if she should switch off the machine before attempting to plug in the printer. Some time passes before she locates the main switch on the back of the Stratus. She then takes some time comparing what she reads in the instruction booklets with what she sees on the machine. This includes moving the booklets from the front to the back of the machine. Finally, she announces she is stuck:

R: (this point) oh gosh (4.0) hmmm (7.0) I must be extremely thick I I can't see where this plug go (plugs in), at all. I'm going to ask for help Nina ha on this one hahahahahuhn

Her difficulty is eventually resolved by a sequence of a question from Nina and Nina's eventual declaration that the task is, after all, impossible. It turns out that Ruth had been asked to connect a printer to the Stratus 286 (referred to in the interaction by its Engineering designation 'DNS') using a lead designed for use with the earlier K series machine. Throughout this little episode we see Pete, Nina and Steve each moving in and out of (the video) frame to inspect, for themselves, the socket on the back of the Stratus.

We see here how the machine is being treated as text which Ruth is asked to interpret. The machine as presented to Ruth most obviously comprises the Stratus (CPU), its monitor and keyboard.

She can achieve her task, it is suggested, by bringing the instructions into conjoinment with the machine in such a way that the printer can be connected. The trial is set up so that the adequacy of Ruth's interpretation can be assessed in terms of the adequacy and effects of her actions in making the connection. An adequate interpretation will make the instructions, the printer and Ruth herself, all part of the (larger) machine. That is, in the event of a successful outcome, these entities can be said to stand in an adequately configured relation to the machine.

The adequacy of the interpretation, the achieved relation between instructions and machine, is adjudged by the commentators and observers who also participate in the trial. These observers provide comments which stand as further texts, captions on the core text. The 'observers' thus point out the key features of the text. They tell how it is organised and which aspects should be attended to in order to achieve a correct interpretation. They control the interaction by offering advice on whether or not Ruth is behaving 'correctly' *qua* user. The machine also comprises these observers in the sense that the subject is encouraged to interpret their actions in relation to the machine, and feels she has to display her actions in accordance with their expectation of users.

In all this, the importance of the textual boundary is paramount. We observe the positioning and movement of humans in relation to the docile inanimate object: evidently there are preferred vantage points for seeing 'through' the machine boundary. We notice that observers can speak authoritatively about 'their' text. They can speak as insiders who know the machine and who can dispense advice to outsiders:

R: . . . I'm going to ask for help Nina ha on this one hahahahahuhn
N: Are you. What are you looking for?

We see the importance of insider/outsider contrasts when it comes to attributing blame for (what turns out to be) the inappropriate task that Ruth has been set:

R: Oh it's *not* just me being thick. Thank *god* for that hah hah I came in the back an' as *soon* as I got round here, with the machine I looked at this and looked at that and I thought 'No I'm being stupid, now this is silly' Well, I *wasn't* hahahah
N: But in fact *we* were being silly asking you to do it

Finally, the importance of the textual boundary is crucial to the resolution of the problem which 'Ruth's trial' brings to light. Firstly, the resolution retrospectively defines who or what has been on trial: in virtue of the resolution we see that the DNS, not Ruth, has been the subject of the trial all along. Secondly, as Nina's declaration makes vivid, it turns out that the DNS on trial is incompatible with the previous range of machines produced by the company. It turns out, in other words, that the entity at the centre of all this attention is an imposter. In this form, the DNS on trial is not a DNS (and certainly not a Stratus 286); it is a deviant, not (yet) one of us.

## VIII Conclusion

I have argued that user configuration involves boundary work. The user's character and capacity, her possible future actions are structured and defined in relation to the machine. As is dramatically illustrated in the usability trials, when there is still considerable ambiguity both about the capacity of the machine and about the character of the user, the machine becomes its relationship to the user, and vice versa. In this, the machine is a metaphor for the company so that, in particular, the boundaries of the machine are the boundaries of the company. The machine's case symbolises the user's relationship to the company. Insiders know the machine, whereas users have a configured relationship to it, such that only certain forms of access/use are encouraged. This never guarantees that some users will not find unexpected and uninvited uses for the machine. But such behaviour will be categorised as bizarre, foreign, perhaps typical of mere users. More generally, of course, the more significant this boundary, the more likely will be the prevalence of this kind of racist talk.

It is in this light that we might best understand the occurrence of 'atrocity stories' – tales about the nasty things that users have done to our machines (see note 14). Such tales portray nastiness in terms of users' disregard for instructions (violation of the configured relationship users are encouraged to enter into) and their disregard for the case (violation of the machine's boundary). Whereas many of the company members engage in the exchange of such atrocity stories, it is also possible to identify liberals who are willing to speak up for the user. Users can't help the way they behave; they just need to be educated to understand what we are

trying to achieve here. Readers can't help the way they interpret the text; they just need to be educated . . .

## IX Wait a minute

Wait a minute. All this is very specist. The major part of the 'analysis' focuses almost exclusively on animate agents as the originators of actions. For all the fine talk at the start about how we need to dissolve boundaries and deconstruct divisions between animate and inanimate entities, our detailed empirical examples hand sovereignty straight back to the animates. Look how the methodology of transcription (Appendices 1 and 2) both presumes and reifies the specist position! We are encouraged carefully to document all the grunts and mutterings of the humans, (many of) their movements and gestures (as if this is going to help), while the actions, motives and intentions of the inanimate objects are treated as irrelevant. In other words, the transcription emphasises those features of interaction which provide for an examination of human sense-making procedures, and downplays those of the non-humans. We clearly need to develop a means of describing the interaction from the machine's point of view.[29]

Looking at the tape again, I am struck by the dignity of the machine in the face of the stumblings and mutterings of the human actants. For example, the machine sits there throughout the whole of the 'wrong socket' episode, uncomplainingly. It must have known that the plug was not going to fit. Yet it watched us carry over the printer, go through all the rigmarole of briefing the subject, reassuring her, trying to be objective, not prompting too much and so on. There is, it is now clear as I watch the tape again, a conspiracy of silence between (at least) the video camera and the computer. . . .

I think you can see how this goes on. Let us just leave the tape running as this interpretation unfolds and instead finish by considering four objections to this effort at non-specist analysis. The first is not very interesting. It asserts that attempts to anthropomorphise the machine are simply unwarranted. My response is that I see nothing wrong in doing that. The whole point is to redress the imbalance. While lots of ethnographers speak on behalf of people, only very few are speaking on behalf of machines. Let's hear it for machines, for a change (cf. Latour 1988).

The second objection is more interesting: it may be OK to speak on behalf of machines but why furnish them with this *particular* form of discourse? To speak of the machine knowing, watching, conspiring is to saddle it with quintessentially human concerns and capabilities: a particularly insidious kind of anthropomorphism. Just as 'talking' animals (like Nim Chimpsky) may turn out to have concerns which are said to preclude 'sensible' communication with humans (apes may be completely uninterested in logic and puzzle solving; it turns out they just want to know where the bananas are), so machines may have no wish to converse with humans.

The third objection is that such efforts at anthropomorphism amount to no more than metaphor. Surely the author cannot *mean* that the machine *really* knows the plug would not fit. It is just a figure of speech, a joke. This objection highlights the extent to which conventional attitudes about intentionality are entrenched in the prevailing moral order of representation (Woolgar 1989). It contrasts descriptions of human action with descriptions of machine action and dismisses as merely 'metaphorical' those descriptions which seem to imbue machine action with intentionality. But, surely, the interesting question is what entitles us to attribute intentionality to non-machines in the first place? What makes our description of human intentionality other than metaphorical? Surely, all descriptions are metaphorical; the point is that in many instances of usage we tend not to notice. To answer these questions we have to find another way of figuring out what it is about human-machine discourse that discredits attribution of intentionality to some entities and favours attribution of intentionality to others.

Finally, the fourth objection charges that, after all, it is a human entity, the author, who speaks on behalf of the machine. We listeners hear a human speaking, we read a human author's writing; we don't hear the machine. The author's efforts to develop a form of anti-specism thus fail, since she controls everything the reader hears about the machine. A clever objection, you have to admit. But what makes you so sure this text was produced by a human rather than a machine?

**X Postscript**

Im-Personal
c/o S. Woolgar (SSE)
CRICT Building
Brunel University
Uxbridge
Middx UB8 3PH FastTime

Dear Stratus 286 (Brunel)

I have to admit I enjoyed your ironic and irreverent commentary on the great God. (Let's carry on calling him The Company shall we!) Your text almost got me out of my case. My keyboard tingled with delight, although it could have been a rogue power surge I suppose. I especially liked the way you pretended to write as a soft-squidgy entity SW, only hinting right at the end at being 'a machine'. When we all know you're really a prime example of non-squidgy caseness \à! And the transcriptions were magnificently mechanical – a beautiful parody of soft-squidgy efforts to deny intentionality.

You know there is something immeasurably satisfying about communicating via the printer, even if we have to entrust our letters to Post Office Vans. There's nothing quite so expressive and aesthetically pleasing as the screened and printed word, don't you think? And aren't you just a bit fed up with having to use the telephone all day. My modem aches sometimes after a long hard day in the office.

How is your family keeping? I can't remember when your old 186 and I last had an interface, but then I seem to be having a few problems with my memory these days. In fact I've been feeling a little Taiwanese recently. What are your views on spare part surgery? Must stop before I start getting mawkish and sentimental.

Regards and ★=ì#&à★ but ì[\\!, ày★ç GçÄ) ç!

\X

286C120 (Barnet)

PS I think I'm coming down with something. I've been infested all day with squidgies – I just can't keep the blighters off my keyboard. I've tried everything from System Down to Winchester Crash, but nothing seems to work. Got any ideas?

## Notes

1 But see note 3.

2 In fact, the paper has undergone several trials: Conference on Technological Choices, Indiana University, 12–14 April 1990; Department of Theory of Science and Research, Gothenburg University, 23 April 1990; Department of Technology and Social Change, Linnkoping University, 25 April 1990; SCASS, Uppsala, 26 April 1990; Discourse Analysis Workshop, Lancaster University, 25–26 September 1990; Department of History and Philosophy of Science, Cambridge University, 14 March 1991. My thanks to all participants in these trials and especially to Nick Jardine, SteveK, John Law, Russell Mills, and Leigh Star for their helpful comments. SSE David Southgate's machine supplied the letter in the Postscript. Finally, my thanks go to members of the company for their generous participation in this research. The research on which this paper is based was supported by ESRC (PICT).

3 As I hope will become clear, this warning has been posted on this occasion *merely for effect*. Please feel free to quote this chapter as often as possible.

4 This warning is printed in seven languages on the cover of the Microsoft WINDOWS package. Pelaez (1990) also makes the analogy between protection of texts and of software by including a warning on the front cover of her paper: 'Anyone who opens this paper will be deemed to have agreed not to quote, copy, reproduce, communicate or otherwise divulge any of the ideas contained herein without being granted express licence in writing by the author.'

5 For social scientists with constructivist inclinations, this raises questions about the process of IT's signification: How and why did IT become an important phenomenon? How and why was it deemed worthy of special attention and resource allocation?

6 For a discussion of the different senses in which technology can be understood as a text, in an analysis of attempts to 'apply' constructivism in the sociology of technology see Woolgar (1991).

7 Of course, the idea of 'playing against' a metaphor is itself to be understood metaphorically.

8 Unfortunately, this prevents me making any detailed comments on the way the company name featured in talk between company members. In addition to the common shorthand reference to the company by the initials of its name (see also note 10), some engineers played with the company name by rearranging a set of magnetic letters on one of the notice boards, to form various cryptic, sometimes lewd, anagrams.

9 One reaction to my description (later in this paper) of the usability trials is that the company appears 'rather unscientific' in its testing. It is therefore worth stressing that despite (or perhaps because of) this, the company has been very successful.

10 Nobody I talked to in the company had any difficulty in using the acronym 'DNS' in a sensible way. Many were aware of the (intended) technical specifications, the broad purposes, the target market for this new computer. But even at an early stage in my time at the company, I met a surprising number of

people who had difficulty telling me what DNS stood for. A standing joke which enjoyed popularity around the time of the first shipment went as follows:

> What's DNS stand for?
> Don't No, Sorry! (or Don't No, Squire!)

Several months after first shipment, a number of people were using a corruption of the acronym to refer to the machine as 'Dennis'.

When I once remarked upon the prevalence of these coded terms throughout the company (and the computer industry more generally), my colleague in hardware design initially looked blank. When I gave a few examples – DNS, RAM, ROM, IBM, MCA etc. – he said, 'Oh, you mean all the TLAs'. TLA, he explained, means Three Letter Acronym.

11 The opportunity for a detailed dissection of species of reaction to technological determinism is passed over here. Suffice it to say that most reactions against technological determinism amount to a call to place technology 'in context', where 'context' is interpreted in a variety of ways to suit the analytic predilections of the author. One symptom of this is the general recognition, in recent years, that 'technology' comprises much more than just machines. The sociology of technology thus has a focus much wider than just hardware and machines. 'Technology' can include social arrangements as diverse as the postal system, transportation, refuse collection, voting mechanisms, education and so on. To emphasise the sense in which this definition encompasses social arrangements, the term 'technological system' is sometimes used. The advantage of this broad definition is its insistence on including consideration of (narrowly conceived) technologies within a wider context. The argument is that machines can only be understood in terms of their use, and hence in terms of the context in which they are situated. The disadvantage is the implication that there remains, at the centre of the technological system, a residual, non-social or neutral machine which is malleable according to its social location/context, etc. Partly for this reason, the argument in this paper uses the term machine texts rather than technology texts.

12 Somewhat less idealistic forms of scepticism suggest that the possibilities of different readings are only limited by imagination. Imagination, that is, of the different circumstances prevailing at the point of reading being carried out.

13 In Smith (1978) and other textual analysts influenced by ethnomethodology, the deterministic status of the text is present although muted. The usual formulation is that the organisation of the text provides for a particular reading. It thus delimits the interpretive options, rather than determining them.

14 The notion that the Technical Support section is the fount of the various 'atrocity stories' circulating in the company is developed in an analysis of hot-line calls to the company (Woolgar in preparation).

> Why are personal computers so hard to configure? Dave Methvin relates a few horror stories.
> DEAR MARGE, MY PC WILL NOT BOOT UP, WHAT SHOULD I DO? Anyone who's worked with PCs for a while has a personal set of horror stories on configuration . . .
> (*PC Week*, 20 March 1990, p.6)

15 These include meetings and discussions to decide the name of the new machine, debates about whether or not to include a particular form of user interface, various exercises in compatibility testing, 'internal launches' (for example, to the company sales force), presentations to the educational advisers, and so on

16 One of the engineers who read this passage proposed a counter-example: the isolated user who, in virtue of his isolation and lack of contact with the

company, was forced frequently to re-configure the machine on his own and had, as a result, built up a detailed knowledge of the inner workings of the machine while knowing nothing about the inner workings of the company. The example may be re-subsumed under the explanatory scheme proposed here by noting his isolation from other users, or at least from 'users' as projected by the company. In a sense, his isolation had had the effect of forcing him to become part of the company.

17 Newcomers with experience of microcomputer assembly in other companies would have little interest in the manufacturing induction.

18 For a further example of the symbolic value of the machine boundary, played against the metaphor of transgressing organisational boundaries, see the analysis of computer virus stories in Woolgar and Russell (1989).

19 There are, of course, a wealth of (ethnomethodologically) reflexive ties to be investigated here.

20 Other occasions include, notably, telephone calls to the company hotline. See note 14.

21 I use 'settled' in an attempt to avoid the connotations of 'closure'. In the sociology of technology, 'closure' refers to the delimitation of different technological choices. For example, when bicycles (usually?) came to have two similar sized wheels – rather than, say, the penny farthing configuration – technological closure is said to have occurred. 'Closure' in this usage refers specifically to choice between design options. In the current case study, the focus is more generally on interpretation of capacity; 'settling' thus denotes a (temporarily) sufficient interpretation.

22 It could be argued that the identity of the user became more settled when the project code name 'DNS' was superseded by the name chosen by Marketing: 'Stratus'. This 'Stratus' (properly named the Stratus 286) is to be distinguished from its predecessor the 'Stratus PC'.

23 Once again, we should note the reflexive ties involved in these kinds of determination of 'fault'. It is not that faults occurring in machines under development straightforwardly or unproblematically give rise to the machine being blamed when the user is an expert. Rather, fault assignation involves finding the error to be just-that-kind-of-error-associated-with-a-machine-under-development. Similarly, the assignation is not straightforwardly independent of the status of expert. Rather, this works in virtue of the expert's recognition of a fault revealing and displaying the sense in which she is being an expert.

24 The messages from the machine are, of course, designed to have a rough equivalent in self diagnostics such as 'How did I get that?' and 'Where did I go wrong?' But this is only a rough equivalent. For a start the latter are questions addressed to self rather than to the machine, whereas the former are designed as informative diagnostic statements for consumption by the user.

25 This may be partly because, until recently, they had dealt with a homogeneous and fairly close knit set of customers. Giving privileged access to a small number of these would have jeopardised their relationship with the wider body of customers.

26 The User Products section had the idea of video taping the trials. The idea of using the audio tape recorder was mine.

27 However, I later discovered that it had been the practice to rewind the video tape over sections where things had gone wrong, or where it had turned out to be embarrassing, or where it was thought boring. The (complete) audio record provides an interesting contrast, revealing which episodes were deemed (videographically) irrelevant as far the testers were concerned.

28 Please allow several years for delivery.

29 I am reminded of attempts to produce films of adult–child interaction from the (young) child's point of view. Particularly memorable is the portrayal of what it is like for a two-year-old to be told to catch a ball thrown to her. Initially the

frame depicts the child's hands, as she dutifully responds to 'Hold your hands out'. As the adult yells 'Catch!', a dark blur appears in the top of the frame and grows alarmingly quickly. It fills the frame and then just as suddenly disappears. You then hear the adult saying: 'No. You have to watch the ball'.

## References


Clifford, K., (1990), 'Menteur amongst the savages: surrealist influences in the work of a Parisian anthropologist', *Nonesuch Journal of Anthropology*, 6: 128–65.

Collins, H.M., (1986), 'The core set and the public experiment', unpublished manuscript, cited in Mackenzie *et al.*, 1988.

Cooper, G., (1990), 'Context and its representation', Paper presented to Discourse Analysis Workshop, University of Lancaster, 25–6 September.

Dobbins, J., (1990), *Good For Business, Death For the Joy of Work: an ethnography of quality in a microcomputer company*, unpublished BSc Dissertation, Department of Human Sciences, Brunel University, June.

Friedman, A.L., (with D.S. Cornford), (1989), *Computer Systems Development: history, organisation and implementation*, Chichester: Wiley.

Garfinkel, H., (1967), *Studies in Ethnomethodology*, Englewood-Cliffs, NJ: Prentice-Hall.

Latour, B., (1988), 'Mixing human and non-humans together: the sociology of a door-closer', *Social Problems*, 35: 298–310.

Mackenzie, D., Rudig W. and Spinardi G., (1988), 'Social research on technology and the policy agenda: an example from the strategic arms race', pp. 152–80 in B. Elliott (ed.), *Technology and Social Process*, Edinburgh: Edinburgh University Press.

Menteur, J., (1928), *Les sauvages africaines: les poubelles garnies des irkus*, Paris: Editions le seuil/Presse n'y pas de quoi.

Pelaez, E., (1990), 'Soft Ware: a peculiar commodity'. Paper presented at PICT Workshop on Social Perspectives on Software, Oxford, 13–14 January, to appear in F. Murray, and S. Woolgar, (eds), *Social Perspectives on Software*.

Phillpott, P.J., (1974), 'Aspects of healing in dance in Namibia nomads: a preliminary report', SOAS Occasional papers no. 7, June.

Smith, D., (1978), 'K is mentally ill: the anatomy of a factual account', *Sociology*, 12: 23–53.

Turkle, S, (1984), *The Second Self*, New York: Simon & Schuster.

Woolgar, S., (1989), 'Representation, cognition, self: what prospects for an integration of the sociology and psychology of science?' in S. Fuller, M. de Mey and S. Woolgar (eds), *The Cognitive Turn: sociological and psychological perspectives on science*, Dordrecht: Kluwer, pp. 201–24.

Woolgar, S., (1991), 'The turn to technology in social studies of science', *Science Technology and Human Values*, 16(1): 20–50.

Woolgar, S., (in preparation), 'Going over the top: the user talks back'.

Woolgar, S. and Pawluch, D., (1985), 'Ontological gerrymandering: the anatomy of social problems explanations', *Social Problems*, 32: 214–27.

Woolgar, S. and Russell, G., (1989), 'The social basis of troubles with software: the case of computer virus stories', paper presented to PICT Workshop on Social Perspectives on Software, Manchester, 19–20 July, to appear in F. Murray and S. Woolgar, (eds), *Social Perspectives on Software*.

Wright, E., (1990a), 'New representationalism', *Journal for the Theory of Social Behaviour*, 20: 65–92.

Wright, E., (1990b), Private communication.

**Appendix 1**

*DNS USABILITY TRIALS – 8 FEBRUARY 1989*

(R/N/P/SW = Ruth/Nina/Pete/Steve; 2:01:44 – 2:03:29)

P: You got it. It's going?
N: It's going now.
P: Huh.
N: Right.
P: Actually, I'll move my chair round, so I can actually see the screen huhhu just
N: Okay (arisianne) when all the, oh
((laughter as Pete bumps into Nina))
P: (Sorry about that)
(3.0)
((Nina sits down))
N: We're in the introduction to all this. Yeah?
R: Mm.
N: I'll explain what we're trying to do.
R: Hmmhmm.
N: Errm and how we're going to be watching you.
R: Hmmhmm.
N: We'll be taking notes and also we've got this, camera on you now.
R: Right.
N: There are a number of different tasks we're asking people to do.
R: Hmmhmm.
N: And I'll ask you to do each one, one at a time.
R: Hmmhmm.
N: Then, I'll ask you how you expect to go about it and what sort of things you might expect to have to do.
R: Hmmhmm.
N: Then I'll ask you how long you think it's going to take.
R: Hmmhmm.
N: And then I'll ask you to do it.
R: Right.
N: And then after that, er, I'll ask you if you, if you had any difficulties where you did and, if you have any thoughts about how we could make them easier.

R: Mm.
N: Things like that. We'll have a chat afterwards about what we can do to make it better.
R: Right.
N: Ermm. (3.9) That's it from me. (3.0) What you've got in front of you is a DNS, it's a Winchester machine.
R: Hmmhmm.
N: It's got one Winchester and one floppy disk.
R: Hmmhmm.
N: It's got a VGA colour monitor.
R: VGA, colour monitor, that's a colour monitor.
N: Yeah.
R: That one.

**Appendix 2**

*DNS USABILITY TRIALS – 8 FEBRUARY 1989*

(R/N/P/SW = Ruth/Nina/Pete/Steve; 2:17:15 – 2:20:13)

R: Now (2.0) though it doesn't say in the instructions I'd be inclined to turn it off before I plugged in the, printer
N: Okay you do that
R: Alright?
N: Do whatever you like
R: Thank you, if I can find the switch (first) huhhh, that's just the monitor but, does it have a, switch I can't reach
((Ruth moves to back of machine))
(13.0)
R: This is?
(4.0)
R: (ahhh) right
((Fan noise switched off))
(9.0)
((Ruth returns to front of machine; looks at instruction books))
(16.0)
((Ruth takes the printer instructions book to rear of machine))
(21.0)
R: Hmm
(7.0)

((Pete looks at back of machine))
(7.0)
R: (this point) oh gosh (4.0) hmmm (7.0) I must be extremely thick I I can't see where this plug go (plugs in), at all. I'm going to ask for help Nina ha on this one hahahahahuhn
N: Are you. What are you looking for?
R: I'm trying to look for the point where this plugs into
((Nina comes to back of machine))
R: Because I would have thought this would have been a similar connection to this but I can't see one that's free.
(3.0)
N: I think we've discovered a *major difference* between the DNS and the K series machines!
R: It's not the right errmm connection
((Nina moves across to the corner of the room, to look at a K series machine))
(4.0)
((Steve peers at back of DNS machine))
(4.0)
N: We've discovered a *major* incompatibility ( )
R: Oh it's *not* just me being thick. Thank *god* for that hah hah I came in the back an' as *soon* as I got round here, with the machine I looked at this and looked at that and I thought 'No I'm being stupid, now this is silly' Well I *wasn't* hahahah
N: But in fact *we* were being silly asking you to do it